

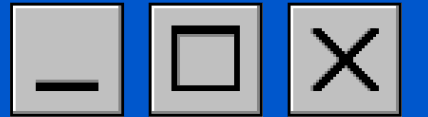
SLIDE 1

DEVICE ARCHITECTURE ANALYSIS

**INTEL CORE I5 (8TH GENERATION) VS.  
APPLE SILICON M4**

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COURSE: COMPUTER ARCHITECTURE CS-391





## SLIDE 2

### APPLE'S M4 CHIP

- ISA: **ARMv9-A (64-bit)**
- ISA Philosophy: **RISC (Reduced Instruction Set Computer)**
- Fixed-length, simpler instructions  
→ higher efficiency

### INTEL I5 8<sup>TH</sup> GEN

- ISA: **x86-64**
- ISA Philosophy: **CISC (Complex Instruction Set Computer)**
- Each instruction capable of performing multiple operations, variable length

### CONTRAST WITH RISC-V

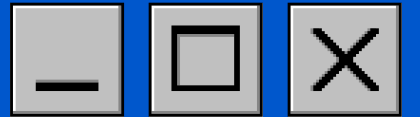
Open-source ISA, No licensing fees  
Used in research and emerging processors



### KEY INSIGHT

ARM (Apple M4) prioritizes efficiency, x86 prioritizes compatibility.





## SLIDE 3

### APPLE'S M4 CHIP

- Very wide superscalar design
- Highly aggressive OoOE
- SIMD: ARM NEON
- AMX (Apple Matrix Extensions for AI/ML)

### INTEL I5 8<sup>TH</sup> GEN

- Superscalar execution
- Out-of-Order Execution (OoOE)
- SIMD: SSE, AVX2

### KEY CONCEPTS

**SIMD:** One instruction operates on many data elements simultaneously

**ILP (Instruction-Level Parallelism):** Multiple instructions at once

**Superscalar:** More than one instruction per cycle

**OoOE:** Instructions reordered to avoid stalls



## SLIDE 4

### APPLE'S M4 CHIP

- Performance cores + Efficiency cores
- No SMT, more physical cores
- Unified LPDDR5X memory

### INTEL I5 8<sup>TH</sup> GEN

- 4-6 cores
- Hyper-Threading (SMT)
- DDR4 RAM

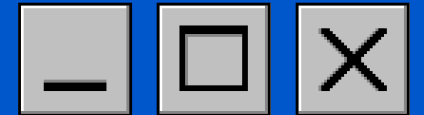
### KEY CONCEPTS

**TLP:** Using multiple cores for parallel tasks, like having multiple brains.

**Unified memory:** In M4, CPU/GPU share the same fast RAM (LPDDR5X), reducing delays.

**Why Apple better?** Hybrid cores (fast for heavy work, efficient for light) + unified memory = better multitasking and speed. Intel uses same-type cores + separate RAM, less optimal.  
No SMT in M4: Relies on real cores instead of virtual ones.





## SLIDE 5

### GEEKBENCH 6 SCORES

#### PERFORMANCE SCORES:

| CPU              | Single-Core | Multi-Core |
|------------------|-------------|------------|
| Intel i5 8th Gen | ~900        | ~3,500     |
| Apple M4         | ~3,800      | ~15,000    |

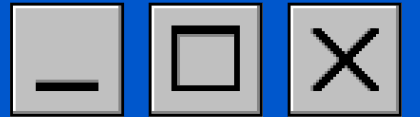


#### WHY APPLE IS BETTER?

- Better ISA efficiency
- More ILP & SIMD
- Better memory system
- 3nm vs 14nm process







## SLIDE 6

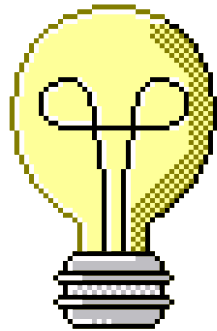
### APPLE'S M4 CHIP

- Built on 3nm fabrication process
- TDP: 10-20W
- Hybrid architecture:
  - Performance cores for heavy tasks
  - Efficiency cores for light tasks
- Aggressive power management:
  - Dynamic Voltage & Frequency Scaling
  - Power gating of unused units

### INTEL I5 8<sup>TH</sup> GEN

- Built on 14nm fabrication process
- TDP: 15-45W
- Homogeneous cores (all cores same type)
- Higher power draw under load
- Performance drops when thermal throttling occurs





### MAIN TAKEAWAYS

- Modern CPU performance comes from architecture, not clock speed
- Parallelism exists at multiple levels: ILP, DLP, and TLP
- Power efficiency is now the primary design goal

### MAJOR STRENGTHS

- Intel Core i5 (8th Gen):
  - Excellent software compatibility
  - SMT (Hyper-Threading) support
- Apple M4:
  - Outstanding performance per watt
  - Unified memory and modern ARM design

### BIGGEST BOTTLENECK

- Intel i5:
  - Aging x86 design and 14nm process
  - Limited efficiency and thermal headroom
- Apple M4:
  - ARM-only software compatibility
  - Closed ecosystem

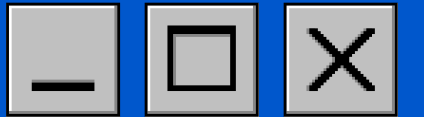


### SURPRISING FINDINGS



- Apple achieves higher performance without SMT
- Massive performance gains are possible at very low power
- Efficiency cores contribute significantly to real-world performance





DO YOU THINK A 'JOHN DOE' FROM ANARKALI BAZAR CAN SPOT PROBLEMS WITH CISC BUT INTEL CANNOT?

 OPEN TO  
QUESTIONS

A pixelated mouse cursor is positioned over the end of the word 'QUESTIONS'.